Ideation Phase
Define the Problem Statements

Date	7 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	**Malaria Detection using Deep Learning**
Maximum Marks	4 Marks

Problem Statement –

Malaria Detection System
Malaria diagnosis is usually done by manually examining blood smear images under a microscope by trained laboratory technicians. This process requires significant time and expertise, and the chances of human error may increase when handling large numbers of samples. In many rural or underdeveloped areas, healthcare facilities lack sufficient diagnostic resources and trained professionals.
To address this issue, an automated system using machine learning and image processing can be developed to detect malaria parasites from blood cell images. This system can assist doctors and laboratory technicians by providing faster and more accurate diagnosis.

Example

Scenario	Traditional Method	Problem Faced	Proposed Solution
A patient visits a hospital with symptoms of malaria	Lab technician examines blood smear under microscope	Diagnosis takes time and requires expert analysis	AI-based system analyzes blood smear images automatically
Large number of patients in a clinic	Manual testing of each sample	Delay in diagnosis and treatment	Automated detection speeds up analysis
Rural healthcare centers	Lack of trained technicians	Incorrect or delayed diagnosis	Machine learning system assists healthcare workers

Expected Outcome

Feature	Benefit
Automated Image Analysis	Faster malaria detection
AI-based Prediction	Accurate results
Web or Mobile Interface	Easy to use for hospitals
Stored Results	Helps maintain patient records